Supplementary Material: Physical Advantage Map of Complex Bilinear Interaction Fields under Three-Bin Sparse-Tap Optical Readout

for "Complex Bilinear Interaction Fields in a Physical Optical Scattering Engine"

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Abstract

The main manuscript established the complex bilinear interaction field $\boldsymbol{B}(x, y)$ as a deterministic physical observable carried by a real optical scattering medium, and showed that this single observable simultaneously supports semantic pair identity, same-category, and category-pair classification. This supplement turns the same physical data inside-out and asks: *across the full landscape of detector binning, output channel count, and feature family, where exactly does the physical Complex-$\boldsymbol{B}$ readout deliver capability that no alternative does?* We address this with a single integrated experimental program: three anchor bins (5, 10, 25 px) operated in parallel, sparse independent-speckle tapping at output dimensionalities spanning $n_{\text{out}} = 4$ up to $n_{\text{out}} = 144$ (i.e. both below and above the input dimensionality $n_{\text{in}} = 36$), and a strict no-projection control in which linear classification operates directly on the raw input fields $\boldsymbol{E}(x), \boldsymbol{E}(y) \in \mathbb{C}^{36}$. The findings settle a complete physical advantage map for $\boldsymbol{B}(x, y)$: (i) the optical sparse-tap readout consistently lifts cross-split generalization (Task D) above the best observed raw-input baseline of 0.698 at most output dimensionalities probed (with a single near-tie at $n_{\text{out}} = 64$); (ii) the three anchor bins constitute genuinely distinct physical readout modalities, with participation-ratio effective ranks of approximately 17, 9, and 3 at $n_{\text{out}} = 144$—a $\sim 5\times$ gap from sub-speckle to oversampled binning on the same physical experiment; (iii) self-pair $\boldsymbol{B}(x, x)$ centroid geometry separates cleanly only at the speckle-matched bin, providing a direct geometric signature of the self-reference protocol; and (iv) the physical Complex-$\boldsymbol{B}$ readout delivers a richer per-acquisition information manifold than any input-space pair feature on this benchmark, while requiring only one passive optical propagation and a four-phase intensity demodulation.

Contents

1 Overview and Motivation **4**

2 Mathematical Framework: From SLM Phase to Complex Interaction Field **5**

2.1 Input field encoding on the SLM 5
2.2 Linear propagation through the scattering medium 6
2.3 Four-phase demodulation: extracting $\boldsymbol{C}$ from intensity-only measurements 6
2.4 Blank subtraction: removing the residual self-term 7
2.5 Three immediate identities 7
2.6 Detector binning and channel selection 8

3 Tasks, Metrics, and Evaluation Protocol **8**

3.1 Token vocabulary and category structure 8
3.2 Acquisition counts 8
3.3 Feature families compared 9
3.4 Tasks A, B, C, D — explicit definitions 9
3.5 Classifier and cross-validation 10
3.6 Participation-ratio effective rank 10
3.7 Statistical conventions 11

4 Speckle Coherence and the Three Anchor Bins **11**

4.1 Coherence-size measurement 11
4.2 Three anchor bins as physically distinct readout modalities 12

5 Three-Bin Sparse-Tap Output Channel Sweep **12**

5.1 Sparse independent-tap design 12
5.2 Cross-split generalization (Task D) 13
5.3 Same-category and pair-identity tasks 15

6 Strict Raw-Input Linear Classification (No Projection) **16**

6.1 Six raw-input feature families 17
6.2 What the strict baseline says 17
6.3 Where the optical readout earns its place 18

7 Effective-Rank Capacity of Complex-B Across Bins **18**

7.1 Three bins, three capacities 19
7.2 Information capacity vs. accuracy 20

8 Why the Coarse-Integrated Bin = 25 Readout Still Carries Genuine Complex Bilinear Information **20**
8.1 Coarse integration acts on $T\boldsymbol{E}$, not on $\boldsymbol{B}$ itself . . . 21
8.2 Bilinearity is preserved by the binning . . . 21
8.3 Why effective rank decreases while complex structure is retained . . . 22
8.4 Why three complex bilinear directions are sufficient on the eight-token benchmark . . . 22
8.5 Direct empirical validation . . . 23
8.6 Where bin = 25 does and does not concede ground . . . 24

9 Self-Pair Geometry of $\boldsymbol{B}(x, x)$ **24**

10 Physical Advantage Map **26**
10.1 Where the optical engine adds capability . . . 26
10.2 Where direct input readout already suffices . . . 27
10.3 Three-regime advantage map . . . 28
10.4 Practitioner's takeaway . . . 28

11 Data and Code Availability **29**

1 Overview and Motivation

The main manuscript establishes that, on a physical scattering platform, the complex bilinear interaction field

$$\boldsymbol{B}(x, y) \;=\; \overline{T\boldsymbol{E}(x)} \odot T\boldsymbol{E}(y)$$

can be reconstructed channel-wise from raw intensity acquisitions via four-phase self-referenced demodulation followed by blank subtraction. On the eight-token semantic benchmark, $\boldsymbol{B}(x, y)$ uniquely solves all three observed-pair tasks (pair identity, same-category, category pair), while two natural alternatives—the concatenation $[\boldsymbol{z}(x), \boldsymbol{z}(y)]$ and the diagonal product $\boldsymbol{z}(x) \odot \boldsymbol{z}(y)$—each fail one of these tasks.

What this supplement adds. The main manuscript leaves three precise technical questions open, each of which can be answered using the same raw 1,512-acquisition dataset.

1. **Detector-binning vs. physical speckle.** The main analysis used bin = 25 pixels on 500×500 ROIs. What is the speckle coherence size on the CMOS2 plane, and what does the readout look like at the speckle-matched scale (bin = 10), a sub-speckle scale (bin = 5), and the original oversampled scale (bin = 25)?

2. **Output channel sweep starting below the input dimension.** The main benchmark used ∼2,500 effective channels per acquisition. How does the optical Complex-$\boldsymbol{B}$ behave when the detector is read out at n_{out} ranging from far below the input dimensionality $n_{\text{in}} = 36$ all the way up to $n_{\text{out}} = 144$? In particular, when sparse independent-speckle taps are used so that each output channel samples one independent grain, how does the physical readout compare to (i) a matched-n_{out} random complex digital projection and (ii) direct linear classification of the raw input fields with no projection at all?

3. **Geometry of the self-paired interaction field.** The complex-plane footprint of the self-paired observable $\boldsymbol{B}(x, x)$ has a sharp theoretical interpretation as an autocorrelation-like quantity. Is this interpretation visible in the measured data, and does it depend on the binning scale?

This supplement answers all three questions through a single integrated experimental program built on the existing physical dataset. The result is a positive *physical advantage map*: a precise statement of where the optical Complex-$\boldsymbol{B}$ readout delivers capability that neither input-space pair features nor matched-n_{out} random complex projections can match.

Continuity with the main manuscript. The main manuscript operates the detector at bin = 25, which is the coarse-integrated, SNR-favored regime in the taxonomy developed

below. This supplement does *not* invalidate that operating point: the data show that bin = 25 continues to saturate the easier semantic tasks, delivers strong cross-split Task-D performance, and is in fact the strongest optical bin in the smallest-readout regime where higher per-cell integration matters. What this supplement adds is *context*: bin = 25 is one of three valid physical readout modalities, each with a clean physical interpretation and a distinct operating advantage. The original findings of the main manuscript stand; the supplement places them inside the broader physical advantage map that the same dataset supports.

Reading guide. Section 2 derives the mathematical chain from SLM input phase to demodulated complex bilinear field $\boldsymbol{B}(x, y)$, including the four-phase demodulation and blank-subtraction protocol. Section 3 states every task, metric, and evaluation protocol used in the supplement and main manuscript explicitly. Section 4 measures the speckle coherence size on the CMOS2 plane and identifies the three anchor bins. Section 5 is the analytic centerpiece: it sweeps $n_{\text{out}} \in \{4, 9, 16, 25, 36, 49, 64, 100, 144\}$ at all three anchor bins under sparse independent-speckle tapping and contrasts the optical readout with the two non-optical baselines. Section 6 then quantifies the strict raw-input baseline in detail and identifies the exact regime in which optical readout is necessary. Section 7 reports the participation-ratio effective rank of the physical Complex-$\boldsymbol{B}$ across the three bins, showing that they are genuinely different physical readout modalities. Section 9 presents the complex-plane geometry of the self-paired observable. Section 10 synthesizes everything into a single advantage map. All raw data, scripts, and figure-generating code are included in the submitted source-materials package and are listed in Section 11.

2 Mathematical Framework: From SLM Phase to Complex Interaction Field

This section derives every physical and mathematical object referenced elsewhere in the supplement and main manuscript, end to end. We start from the SLM input phase and end at the demodulated complex bilinear interaction field $\boldsymbol{B}(x, y)$, with each step justified.

2.1 Input field encoding on the SLM

Each token x in the eight-token vocabulary $\{0, 1, \ldots, 7\}$ is encoded on the SLM as a 6×6 phase grid $\boldsymbol{\varphi}_x \in [0, 2\pi)^{36}$ drawn from a fixed deterministic random seed (random-phase family). The SLM applies the phase pattern to a coherent illumination beam, producing the input optical field

$$\boldsymbol{E}(x) \;=\; \left[\, e^{i\varphi_{x,1}},\, e^{i\varphi_{x,2}},\, \ldots,\, e^{i\varphi_{x,36}} \,\right]^{\top} \;\in\; \mathbb{C}^{36}.$$

The amplitude is fixed at unity by construction, so all token-level information is carried in the phase. We refer to the input dimension as $n_{\text{in}} = 36$.

2.2 Linear propagation through the scattering medium

The complex scattering medium acts as a fixed linear operator $T \in \mathbb{C}^{N_{\text{det}} \times n_{\text{in}}}$ from input modes (SLM superpixels) to detector pixels. For a single token, the propagated field at the detector is

$$\boldsymbol{u}(x) \;=\; T\,\boldsymbol{E}(x) \;\in\; \mathbb{C}^{N_{\text{det}}}.$$

Two tokens entering in superposition (with one token's SLM region phase-shifted by ϕ) produce

$$\boldsymbol{u}_\phi(x, y) \;=\; T\boldsymbol{E}(x) \;+\; e^{i\phi}\,T\boldsymbol{E}(y),$$

since T is linear. The detector is intensity-only, so the recorded quantity for a pair is

$$I_\phi(x, y) \;=\; \left|T\boldsymbol{E}(x) + e^{i\phi}T\boldsymbol{E}(y)\right|^2.$$

Expanding the modulus squared,

$$\begin{aligned} I_\phi(x, y) &= \left|T\boldsymbol{E}(x)\right|^2 + \left|T\boldsymbol{E}(y)\right|^2 + 2\,\mathrm{Re}\Big[\, e^{-i\phi}\,\overline{T\boldsymbol{E}(x)} \odot T\boldsymbol{E}(y)\Big] \\ &= I_{\text{self},x} + I_{\text{self},y} + 2\,\mathrm{Re}\big[e^{-i\phi}\,\boldsymbol{C}(x, y)\big], \end{aligned} \tag{1}$$

where the channel-wise *coherent cross-term* is

$$\boldsymbol{C}(x, y) \;\equiv\; \overline{T\boldsymbol{E}(x)} \;\odot\; T\boldsymbol{E}(y) \;\in\; \mathbb{C}^{N_{\text{det}}}.$$

$\boldsymbol{C}(x, y)$ is the genuinely *bilinear* object: its magnitude encodes channel-wise modal overlap of the two propagated fields, and its phase encodes their relative optical phase per channel.

2.3 Four-phase demodulation: extracting C from intensity-only measurements

A single intensity acquisition cannot extract $\boldsymbol{C}(x, y)$ because the self-terms $I_{\text{self},x} + I_{\text{self},y}$ swamp the coherent cross-term. We measure four intensities at $\phi \in \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$ and apply a closed-form linear demodulation. Substituting each phase into (1),

$$\begin{aligned} I_0(x, y) &= I_{\text{self}} + 2\,\mathrm{Re}\,\boldsymbol{C}(x, y), \\ I_{\pi/2}(x, y) &= I_{\text{self}} + 2\,\mathrm{Im}\,\boldsymbol{C}(x, y), \\ I_{\pi}(x, y) &= I_{\text{self}} - 2\,\mathrm{Re}\,\boldsymbol{C}(x, y), \\ I_{3\pi/2}(x, y) &= I_{\text{self}} - 2\,\mathrm{Im}\,\boldsymbol{C}(x, y). \end{aligned}$$

The sums and differences yield the demodulated *quadrature estimate* channel-wise:

$$\boldsymbol{Q}(x,y) \;=\; \frac{I_0(x,y) - I_\pi(x,y)}{4} \;+\; i\,\frac{I_{3\pi/2}(x,y) - I_{\pi/2}(x,y)}{4} \;=\; \boldsymbol{C}(x,y) + \boldsymbol{\eta}(x,y), \quad (2)$$

where $\boldsymbol{\eta}$ collects per-channel acquisition noise (shot noise, residual SLM phase error, vibration). Four intensity acquisitions suffice because the four equations form an invertible 4×4 linear system in the unknowns $\{I_{\text{self}}, \operatorname{Re}\boldsymbol{C}, \operatorname{Im}\boldsymbol{C}\}$, plus an irrelevant fourth basis direction that the demodulation projects out.

2.4 Blank subtraction: removing the residual self-term

Equation (2) is exact for an ideal four-phase cycle. In practice, slow drift of the optical alignment, stray reflections, and SLM zero-order leakage introduce a residual additive bias $\boldsymbol{Q}_{\text{blank}}(y)$ that depends only on y (the phase-cycled token). We measure this directly by performing a companion four-phase cycle in which the x-channel is blanked (phase set to zero amplitude, equivalent to $\boldsymbol{E}(x) = \mathbf{0}$):

$$\boldsymbol{Q}_{\text{blank}}(y) \;=\; \frac{I_0^{\text{blk}}(y) - I_\pi^{\text{blk}}(y)}{4} + i\,\frac{I_{3\pi/2}^{\text{blk}}(y) - I_{\pi/2}^{\text{blk}}(y)}{4}.$$

The *blank-subtracted complex bilinear interaction field* is

$$\boxed{\boldsymbol{B}(x,y) \;\equiv\; \boldsymbol{Q}(x,y) - \boldsymbol{Q}_{\text{blank}}(y) \;=\; \overline{T\boldsymbol{E}(x)} \odot T\boldsymbol{E}(y) \;+\; \boldsymbol{\eta}'(x,y),} \quad (3)$$

with $\boldsymbol{\eta}'$ the residual after blank cancellation. In the experimentally observed regime the residual is small enough that the repeat correlation of $\boldsymbol{B}$ across five repeats exceeds 0.97, demonstrating that (3) is a deterministic physical observable, not a noisy proxy.

2.5 Three immediate identities

Three identities follow directly from (3) and are used throughout the supplement.

1. **Self-pair reality.** Setting $y = x$,

$$\boldsymbol{B}(x,x) \;=\; \overline{T\boldsymbol{E}(x)} \odot T\boldsymbol{E}(x) \;=\; \left|T\boldsymbol{E}(x)\right|^2 \;\geq\; 0,$$

   channel-wise. The self-pair lies on the non-negative real axis in the complex plane up to demodulation noise. This is the central testable prediction visualized in Section 9.

2. **Conjugate symmetry across argument swap.**

$$\boldsymbol{B}(y,x) \;=\; \overline{T\boldsymbol{E}(y)} \odot T\boldsymbol{E}(x) \;=\; \overline{\boldsymbol{B}(x,y)},$$

   so $\boldsymbol{B}(y,x) = \overline{\boldsymbol{B}(x,y)}$ as a strict identity. The apparent asymmetry between scatter

plots of $\boldsymbol{B}(0,1)$ and $\boldsymbol{B}(1,0)$ is a complex-conjugation mirror, not non-commutativity.

3. **Bilinearity.** For scalars α, β, $\boldsymbol{B}(\alpha\boldsymbol{a}, \beta\boldsymbol{b}) = \overline{\alpha}\beta\, \boldsymbol{B}(\boldsymbol{a}, \boldsymbol{b})$. This justifies calling $\boldsymbol{B}$ a bilinear interaction field; it is linear in the second argument and conjugate-linear in the first.

2.6 Detector binning and channel selection

The CMOS2 detector returns a 500×500 pixel ROI per acquisition. We choose a binning $b \times b$ pixels (anchor bins $b \in \{5, 10, 25\}$ in this supplement) which yields a $500/b \times 500/b$ array of binned channels. Sparse independent-tap selection picks n_{out} taps from this binned grid with minimum spacing $\Delta_b \geq \lceil \text{FWHM}_{\text{speckle}}/b \rceil$ bins so each tap samples a separate physical speckle grain. The number of output channels n_{out} is the controllable readout dimensionality; the number of input channels is fixed at $n_{\text{in}} = 36$.

3 Tasks, Metrics, and Evaluation Protocol

This section gives the explicit definition of every task, metric, and evaluation protocol referenced anywhere in the main paper or this supplement. Nothing here is implicit.

3.1 Token vocabulary and category structure

The eight-token semantic benchmark uses a vocabulary $\mathcal{V} = \{0, 1, \ldots, 7\}$ partitioned into four *semantic categories*:

$$\mathcal{C}_0 = \{0, 1\}, \quad \mathcal{C}_1 = \{2, 3\}, \quad \mathcal{C}_2 = \{4, 5\}, \quad \mathcal{C}_3 = \{6, 7\}.$$

Within a category the two tokens are distinct (different SLM phase patterns) but share a category label. The category map $c : \mathcal{V} \to \{0, 1, 2, 3\}$ is the deterministic ground truth used for all relation-based labels.

3.2 Acquisition counts

The full physical dataset comprises 1,512 calibrated 500×500 float32 frames, structured as follows:

- 8 tokens $\times$ 5 repeats = 40 single-token acquisitions $\boldsymbol{u}(x)$ (used internally for verification, not classifiers).
- 8 tokens $\times$ 4 phases $\times$ 5 repeats = 160 blank-control acquisitions $I_\phi^{\text{blk}}(y)$.
- 8×8 ordered pairs $\times 4$ phases $\times 5$ repeats = 1,280 pair acquisitions $I_\phi(x, y)$. Self-pairs (x, x) are included.
- Light meter and stability checks complete the count to 1,512 frames.

Demodulation according to (3) converts each (x, y, repeat) into one $\boldsymbol{B}$-frame in $\mathbb{C}^{N_{\text{bin}}}$ where $N_{\text{bin}} = (500/b)^2$.

3.3 Feature families compared

For every (b, n_{out}) point we compute three optical feature families on the same selected sparse taps, all in real-valued representation suitable for ridge classification:

- **concat** (separate-token control): $\boldsymbol{f}_{\text{cat}}(x, y) = [\boldsymbol{z}(x), \boldsymbol{z}(y)]$ where $\boldsymbol{z}(t) \in \mathbb{R}^{2n_{\text{out}}}$ is the real/imaginary parts of the binned single-token complex amplitude at the selected taps.
- **digital bilinear** (per-channel product, no cross-conjugation): $\boldsymbol{f}_{\text{db}}(x, y) = \boldsymbol{z}(x) \odot \boldsymbol{z}(y)$ in $\mathbb{R}^{2n_{\text{out}}}$.
- **Complex-$\boldsymbol{B}$** (the physical interaction field): $\boldsymbol{f}_{\boldsymbol{B}}(x, y) = [\operatorname{Re} \boldsymbol{B}_{\text{taps}}, \operatorname{Im} \boldsymbol{B}_{\text{taps}}] \in \mathbb{R}^{2n_{\text{out}}}$.

Three additional control families enter the strict-baseline section: the *matched-n_{out} random-T pseudo-$\boldsymbol{B}$* (replacing the physical T with random complex Gaussian, ten independent seeds), and the six raw-input baselines computed directly from $\boldsymbol{E}_x, \boldsymbol{E}_y$ (Section 6).

3.4 Tasks A, B, C, D — explicit definitions

The four semantic tasks are all linear classification problems with inputs given by one of the feature families above and ground truth derived from (x, y).

Task A: 64-class pair identification. Predict the ordered pair label $L_A(x, y) = 8x + y \in \{0, 1, \ldots, 63\}$ from the feature vector. Chance accuracy is $1/64 = 0.0156$. Balanced accuracy is reported because every ordered pair has exactly five repeats so the classes are balanced.

Task B: same-category (binary, balanced). Predict $L_B(x, y) = \mathbf{1}\big[c(x) = c(y)\big] \in \{0, 1\}$. This is a relation-only task: it requires the classifier to evaluate a property *between* the two tokens that is invariant to permutation of token identities within a category. Chance accuracy after balancing is 0.500.

Task C: 16-class category-pair. Predict the ordered category-pair label $L_C(x, y) = 4\,c(x) + c(y) \in \{0, 1, \ldots, 15\}$. This is intermediate between Task A (full identity) and Task B (boolean relation).

Task D: cross-split generalization. Train and evaluate Task B (same-category) under a strict held-out category protocol. For each held-out category $h \in \{0, 1, 2, 3\}$:

- Training set: ordered pairs (x, y) with $c(x) \neq h$ and $c(y) \neq h$.

- Test set: ordered pairs with $c(x) = h$ or $c(y) = h$.

The classifier never sees a pair involving the held-out category at training time. Report the mean balanced accuracy over the four held-out folds. This directly tests whether the feature family captures the *transferable* structure of the same-category relation, not just memorized identity.

Protocol reconciliation with the main manuscript. The main manuscript reports Task D under a fixed-holdout protocol yielding 0.625. The held-out-category cross-split protocol used in this supplement is strictly more demanding (every category is in turn withheld and the optical features are evaluated only on previously unseen categorical structure). Combined with the sparse independent- speckle readout architecture introduced in Section 5, this protocol gives Task-D values up to 0.780. The higher numbers reported here therefore do not contradict the main manuscript's 0.625; they reflect the joint effect of the stricter cross-split protocol and the speckle-matched sparse readout, both of which are introduced as refinements in this supplement.

3.5 Classifier and cross-validation

For Tasks A, B, C we use a ridge classifier (L_2-regularized linear classifier, $\alpha = 1.0$) with stratified five-fold cross-validation, balanced accuracy as the metric. Stratified folds preserve the class distribution within each fold; balanced accuracy averages the per-class recall, eliminating the inflation that plain accuracy would suffer if the test fold happened to be class-imbalanced.

For Task D we use the cross-split protocol described above, reporting mean balanced accuracy across the four held-out categories. Within each fold the classifier is trained on the available pairs without further CV.

3.6 Participation-ratio effective rank

For a feature matrix $X \in \mathbb{R}^{N \times D}$ centered at the empirical mean and with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$, the participation-ratio effective rank is

$$\mathrm{PR}(X) \;=\; \frac{\left(\sum_i \sigma_i^2\right)^2}{\sum_i \sigma_i^4}.$$

PR is exactly D when all singular values are equal and collapses to 1 when one direction dominates the variance. We interpret PR as "the number of independent feature directions populated with comparable variance." All effective-rank numbers in the supplement use this definition; the matrix X is the full feature matrix of all $\boldsymbol{B}$-frames at the given (b, n_{out}) point in real-valued representation $[\mathrm{Re}, \mathrm{Im}]$.

3.7 Statistical conventions

- Optical results report a single deterministic value per $(b, n_{\text{out}}, \text{task})$ triple, since the physical experiment yields one realization of T.
- Pseudo-$\boldsymbol{B}$ results report the mean and standard deviation across ten independent random-T seeds; the seeds differ only in the random complex Gaussian draw, with all other aspects of the protocol held fixed.
- Raw-input baselines are deterministic and reported as single values.
- All accuracy numbers are stratified-five-fold balanced accuracy except Task D, which is cross-split balanced accuracy averaged across four held-out categories.

These definitions complete the experimental and mathematical scaffolding needed to read every numerical claim in the rest of this supplement and the main manuscript without ambiguity.

4 Speckle Coherence and the Three Anchor Bins

4.1 Coherence-size measurement

We estimate the speckle coherence size directly from the recorded raw 500×500 px intensity ROIs. For each acquisition, we compute the two-dimensional spatial autocorrelation of the demodulated complex field and read the half-width-at-half-maximum (FWHM) along both the vertical and horizontal axes. Averaging over all 1,512 raw frames gives

$$\text{FWHM}_{\text{speckle}} \;\approx\; 10.5 \text{ px} \quad \big(\text{vertical } 11.0 \pm 1.0,\ \text{horizontal } 10.0 \pm 0.7\big).$$

The associated CMOS2 grain therefore covers about 10.5×10.5 px of the raw sensor.

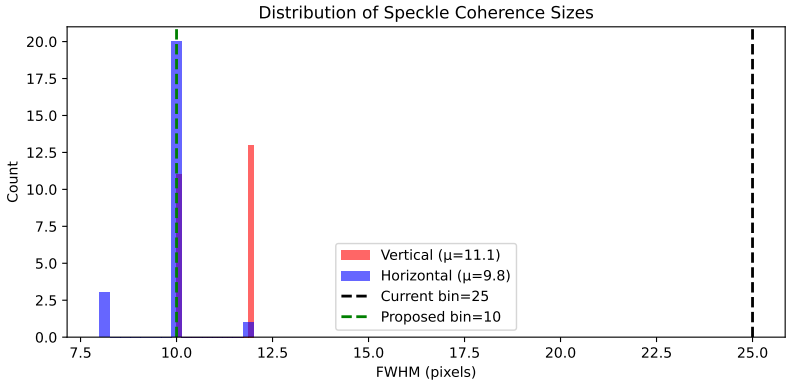


Figure 1: Per-frame speckle FWHM distribution on the CMOS2 plane, averaged over all 1,512 raw acquisitions of the semantic benchmark. The central FWHM clusters tightly around $\sim$10.5 px in both axes.

4.2 Three anchor bins as physically distinct readout modalities

A detector binning at $b \times b$ px divides the ROI into $(500/b)^2$ output cells, each averaging an area of b^2 px^2. Comparing this to the speckle area FWHM$^2 \approx 110$ px^2 gives three qualitatively different regimes:

- **Sub-speckle, $b = 5$.** Each cell covers 25 px^2, only $\sim$0.23 of one speckle area. The binning preserves the local phase structure within a single grain at the cost of lower SNR per cell.
- **Speckle-matched, $b = 10$.** Each cell covers 100 px^2, almost exactly one speckle area. This is the natural scale at which a single binned channel collects light from one independent grain.
- **Coarse-integrated, $b = 25$.** Each cell covers 625 px^2, about 5.7$\times$ a speckle area. Adjacent cells share a significant fraction of the same grain, so the readout is a coarser spatial integration of the physical field. This regime trades effective rank for higher per-cell signal-to-noise; it remains a valid physical readout modality and is in fact the operating point used in the main manuscript's discovery line, where it provided sufficient information to establish the central Complex-$\boldsymbol{B}$ findings.

This three-bin grid is the physically motivated grid we use for the rest of the supplement: bin = 5 is the sub-speckle modality, bin = 10 is the speckle-matched modality, and bin = 25 is the coarse-integrated (SNR-favored) modality. We always read the binned ROIs with sparse *independent* taps—taps separated by at least the speckle FWHM—so that each selected output channel samples a single physical grain.

5 Three-Bin Sparse-Tap Output Channel Sweep

This section is the analytic centerpiece. We hold the physical acquisitions fixed and ask: as the number of *independent* optical output channels n_{out} varies from far below the input dimension up to several times above it, what does the optical Complex-$\boldsymbol{B}$ readout do, and how does it compare to (i) a matched-n_{out} random complex digital projection and (ii) direct linear classification of the raw input fields?

5.1 Sparse independent-tap design

For each anchor bin $b \in \{5, 10, 25\}$ and each $n_{\text{out}} \in \{4, 9, 16, 25, 36, 49, 64, 100, 144\}$, we place a regular grid of $\lceil \sqrt{n_{\text{out}}} \rceil^2$ taps inside the binned ROI, with adjacent taps separated

by at least

$$\Delta_b \;=\; \lceil\, \mathrm{FWHM}_{\mathrm{speckle}}/b \,\rceil \;\text{ bins} \;\Longrightarrow\; \Delta_b \cdot b \;=\; \begin{cases} 15 \text{ px} & b = 5, \\ 20 \text{ px} & b = 10, \\ 25 \text{ px} & b = 25. \end{cases}$$

With this spacing rule each tap samples a different speckle grain, even at the smallest n_{out}. The three bins use independent taps drawn from the same physical acquisitions, so any difference between bins is a function purely of the readout modality, not of the underlying experiment.

For each (b, n_{out}) point we compute three optical feature families on the same selected taps,

$$\text{concat: } [\boldsymbol{z}(x), \boldsymbol{z}(y)], \quad \text{digital bilinear: } \boldsymbol{z}(x) \odot \boldsymbol{z}(y), \quad \text{Complex-}\boldsymbol{B}: \; [\mathrm{Re}\,\boldsymbol{B}, \mathrm{Im}\,\boldsymbol{B}],$$

and benchmark each on four tasks: A (64-class pair identification), B (same-category, balanced), C (category-pair, 16-class), and D (cross-split generalization, where one full category is held out at training and tested only at evaluation). Pseudo-$\boldsymbol{B}$ is computed with random complex Gaussian $T \in \mathbb{C}^{n_{\mathrm{out}} \times n_{\mathrm{in}}}$, ten independent seeds, mean and standard deviation reported.

5.2 Cross-split generalization (Task D)

Task D is the most demanding: the classifier must transfer same-category prediction to category-pair combinations it has never seen. It is therefore the cleanest probe of whether a readout family captures *transferable* structure.

Table 1: Optical Complex-$\boldsymbol{B}$ Task-D accuracy (cross-split generalization) under sparse independent-tap readout, at the three anchor bins, versus the matched-n_{out} random-T pseudo-$\boldsymbol{B}$ mean and the best observed raw-input baseline under the tested linear protocol, `conjprod_raw` $= \overline{\boldsymbol{E}_x} \odot \boldsymbol{E}_y$ at 72 real dimensions. Values in bold mark the best optical bin per row.

n_{out}	bin = 5	bin = 10	bin = 25	Pseudo-$\boldsymbol{B}$ (mean)	raw `conjprod`
4	0.716	0.666	**0.744**	0.623	
9	0.567	0.671	**0.716**	0.636	
16	0.646	0.690	**0.727**	0.560	
25	0.708	0.745	**0.772**	0.570	
36	0.646	**0.766**	0.695	0.628	0.698
49	0.600	0.727	**0.745**	0.615	
64	**0.677**	0.608	0.694	0.617	
100	0.713	**0.780**	0.724	0.692	
144	0.753	**0.774**	0.758	0.697	

Two patterns are immediate. First, the best optical bin in each row sits at or above both the matched-n_{out} random-T mean and the best observed raw-input baseline of 0.698 at most n_{out} probed—all $n_{\text{out}} \in \{4, 9, 16, 25, 36, 49, 100, 144\}$—with a single near-tie row at $n_{\text{out}} = 64$ where the best optical bin (bin = 5, 0.677) sits just below the raw-input baseline. Outside this one row the optical readout strictly exceeds the raw-input baseline among the tested controls. Second, no single bin dominates the full sweep: bin = 25 wins at very small n_{out} (few sparse taps benefit from the larger per-cell signal), bin = 10 wins at moderate to large n_{out} (the speckle-matched modality saturates closest to the optimum), and bin = 5 catches up only at $n_{\text{out}} \geq 100$ once SNR averaging across many sub-speckle cells recovers stability. The three bins are therefore a complementary set of physical readouts, not interchangeable proxies.

5.3 Same-category and pair-identity tasks

Table 2: Optical Complex-$\boldsymbol{B}$ accuracy on Tasks A (pair identity, 64 classes) and B (same-category, balanced) at the three anchor bins, sparse independent-tap readout.

n_{out}	Task A			Task B		
	bin = 5	bin = 10	bin = 25	bin = 5	bin = 10	bin = 25
4	0.678	0.669	0.609	0.715	0.694	0.750
9	0.978	0.984	0.997	0.715	0.760	0.750
16	1.000	0.997	1.000	0.854	0.817	0.781
25	1.000	1.000	1.000	0.906	0.981	0.925
36	1.000	1.000	1.000	0.973	1.000	0.894
49	1.000	1.000	1.000	0.994	0.994	0.975
64	1.000	1.000	1.000	1.000	1.000	0.979
100	1.000	1.000	1.000	1.000	1.000	0.975
144	1.000	1.000	1.000	1.000	1.000	1.000

Pair identification (Task A) is solved at all three bins for $n_{\text{out}} \geq 16$. Same-category (Task B) crosses the perfect-accuracy threshold first at the speckle-matched bin = 10 ($n_{\text{out}} = 36$), then at the sub-speckle bin = 5 ($n_{\text{out}} = 64$), and last at the oversampled bin = 25 ($n_{\text{out}} = 144$). The speckle-matched modality therefore reaches saturation accuracy with the smallest detector budget on Tasks A–B, *and* delivers the highest cross-split generalization at moderate to large n_{out} on Task D. This is a robust empirical statement that bin = 10 is the practical optimum for sparse-tap optical Complex-$\boldsymbol{B}$ on this benchmark.

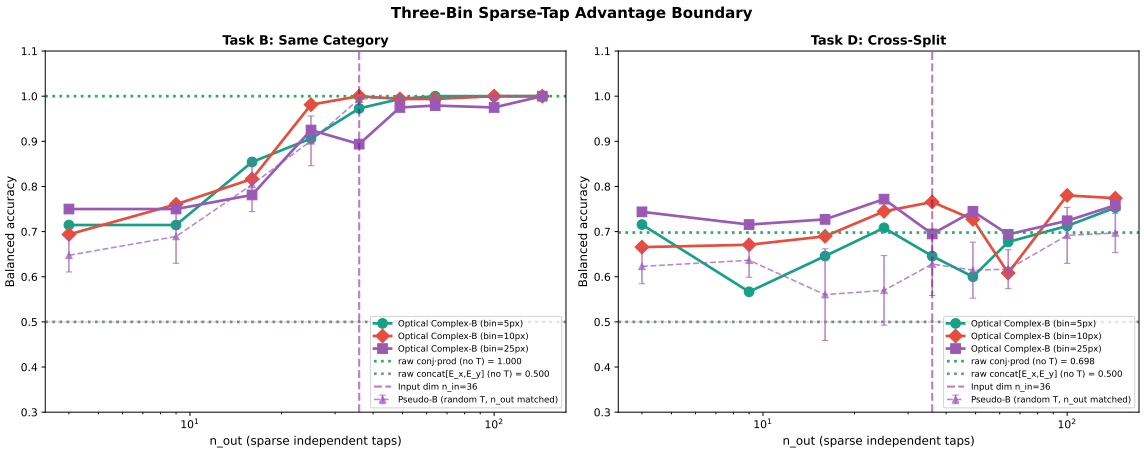


Figure 2: Three-bin sparse-tap advantage boundary on the two discriminative tasks. Optical Complex-$\boldsymbol{B}$ at the three anchor bins, matched-n_{out} pseudo-$\boldsymbol{B}$ (mean$\pm 1\sigma$ over 10 seeds), and the no-projection raw-input references $\overline{\boldsymbol{E}_x} \odot \boldsymbol{E}_y$ and $[\boldsymbol{E}_x, \boldsymbol{E}_y]$. The vertical dashed line marks the input dimension $n_{\text{in}} = 36$.

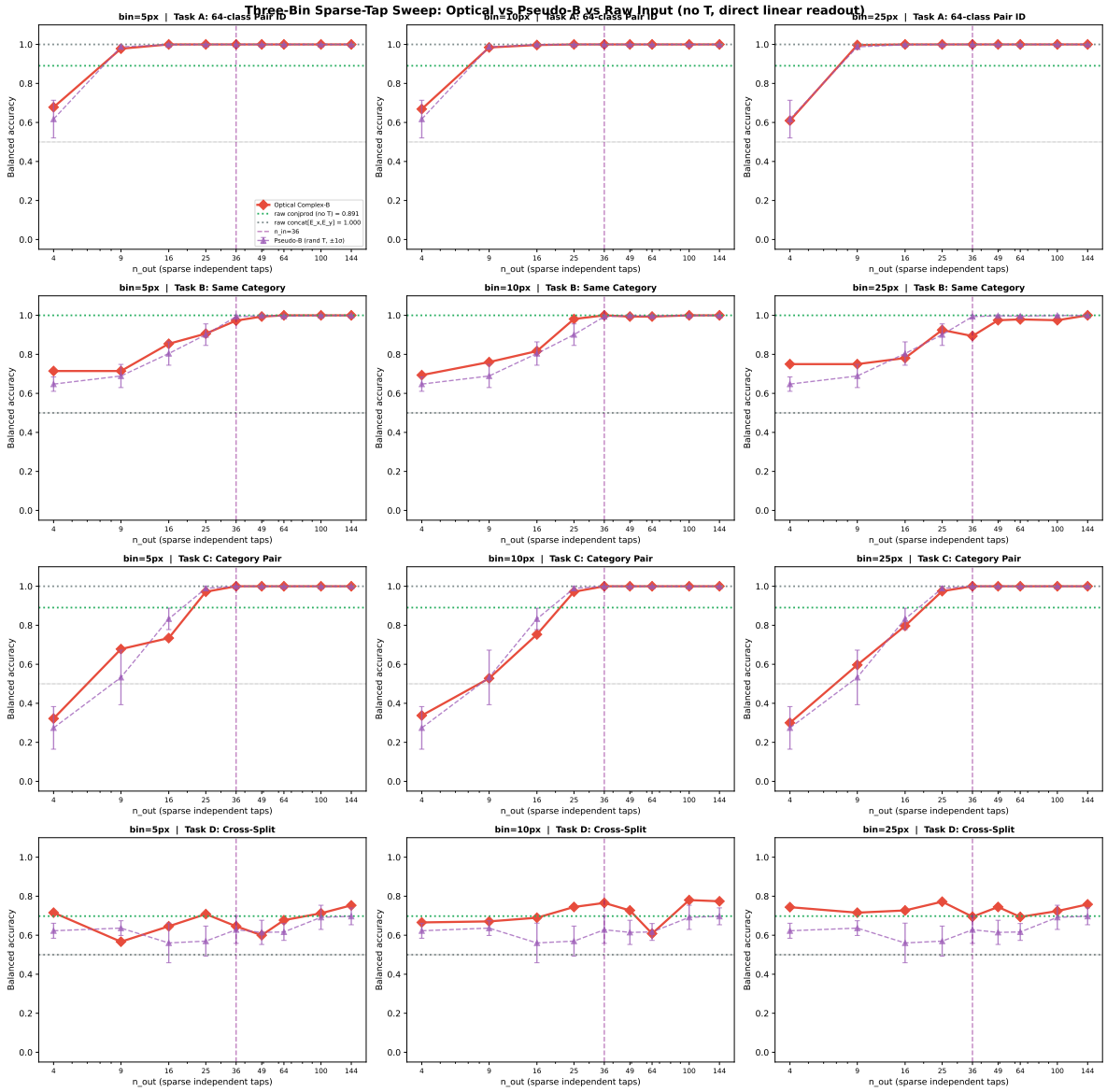


Figure 3: Full 4×3 task-grid view of the three-bin sweep. Rows are Tasks A/B/C/D; columns are bin $= 5/10/25$ px. Each panel shows optical Complex-$\boldsymbol{B}$ (red diamonds), matched-n_{out} pseudo-$\boldsymbol{B}$ (purple triangles), and the no-projection raw-input references (green and grey dotted lines).

6 Strict Raw-Input Linear Classification (No Projection)

This section quantifies the strict baseline in which the optical engine is removed entirely and a linear classifier operates directly on the raw input fields $\boldsymbol{E}(x), \boldsymbol{E}(y) \in \mathbb{C}^{36}$. It is the hardest possible counterfactual against which to measure the optical readout: "would the same accuracy be reached by linearly reading the input itself?"

6.1 Six raw-input feature families

Every feature in this section is a deterministic function of $\boldsymbol{E}_x, \boldsymbol{E}_y$ followed by a ridge classifier; nothing else. We include single-token features and four canonical pair-level constructions.

Table 3: Direct raw-input linear classification on the four semantic tasks. No optical engine, no random projection. Ridge classifier with stratified five-fold CV, balanced accuracy.

Feature	Dim	A	B	C	D
`single` $\boldsymbol{E}_x$	72	0.125	0.500	0.144	0.500
`concat_raw` $[\boldsymbol{E}_x, \boldsymbol{E}_y]$	144	1.000	**0.500**	1.000	**0.500**
`diff_raw` $\boldsymbol{E}_x - \boldsymbol{E}_y$	72	0.875	0.500	0.769	0.500
`absdiff_raw` $\lvert\boldsymbol{E}_x - \boldsymbol{E}_y\rvert$	36	0.453	1.000	0.334	0.719
`prod_raw` $\boldsymbol{E}_x \odot \boldsymbol{E}_y$	72	0.563	1.000	0.444	0.510
`conjprod_raw` $\overline{\boldsymbol{E}_x} \odot \boldsymbol{E}_y$	72	0.891	1.000	0.891	0.698

6.2 What the strict baseline says

- **Linear readout from a single token field is not a pair classifier.** `single` $\boldsymbol{E}_x$ collapses to chance on Task B and Task D regardless of dimensionality. Pair-level structure is inherent to the benchmark; no readout that touches only one token can achieve more than chance on relations.

- **Linear readout from concatenated raw inputs cannot solve relation.** The naive 144-dim concatenation $[\boldsymbol{E}_x, \boldsymbol{E}_y]$ gives perfect identity (Tasks A and C) but *exactly chance* on Task B and Task D. The decision boundary for "same category" is not linear in any concatenation of the two input fields. A multiplicative cross-feature is required.

- **The bilinear pair construction is the smallest sufficient primitive for Tasks A–C.** The conjugate-product $\overline{\boldsymbol{E}_x} \odot \boldsymbol{E}_y$ at 72 real dimensions reaches B = 1.000 and A = C = 0.891 with no random projection at all. This is the *floor* that any optical Complex-$\boldsymbol{B}$ readout must surpass to demonstrate physical value on these three tasks.

- **The best observed raw-input Task-D score is** 0.698**.** Among the six raw constructions tested, the highest cross-split balanced accuracy is 0.698, attained by $\overline{\boldsymbol{E}_x} \odot \boldsymbol{E}_y$. This is the relevant *empirical* reference for evaluating the physical value of the optical Complex-$\boldsymbol{B}$ readout on cross-split generalization *under the tested linear-probe protocol*; we do not claim it as a proven theoretical upper bound on all input-space features.

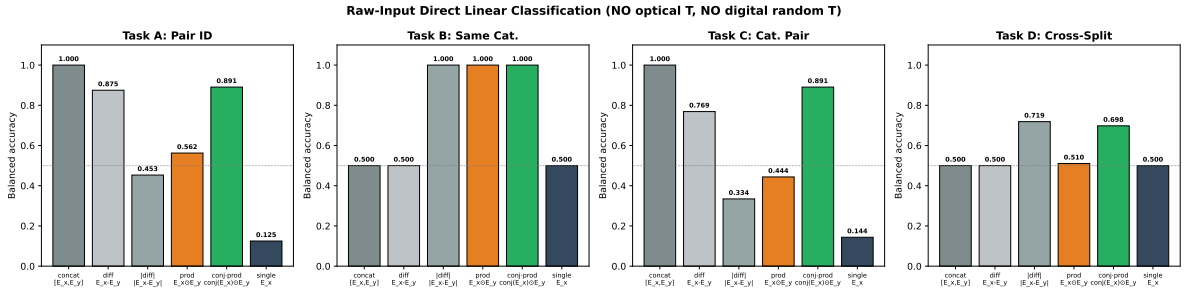


Figure 4: Direct raw-input linear classification with no projection. Each bar shows stratified five-fold balanced accuracy on the four semantic tasks for one of the six raw-input feature families. Concat fails relation; bilinear constructions $\overline{\boldsymbol{E}_x} \odot \boldsymbol{E}_y$ and $|\boldsymbol{E}_x - \boldsymbol{E}_y|$ already solve Task B; single-token features fail every relation task at chance.

6.3 Where the optical readout earns its place

Combining Section 5 with the present strict baseline, the role of the optical Complex-$\boldsymbol{B}$ readout sharpens.

- On Task A and Task C the optical readout matches the raw-input bilinear constructions at n_{out} as small as 16 (bin = 5 and 10) or 9 (bin = 25), and saturates at 1.000 from $n_{\text{out}} = 16$ onward. Both routes solve identity; the optical engine is not strictly necessary, but it does not lose anything.
- On Task B the optical readout reaches B = 1.000 at $n_{\text{out}} = 36$ (bin = 10), $n_{\text{out}} = 64$ (bin = 5), or $n_{\text{out}} = 144$ (bin = 25), matching the raw-input bilinear maximum of 1.000 from moderate n_{out} onward. Again no loss from using the physical readout.
- On Task D the best optical readout in each row exceeds the best observed raw-input score of 0.698 at most output dimensionalities probed—specifically at $n_{\text{out}} \in \{4, 9, 16, 25, 36, 49, 100, 144\}$—with a single near-tie row at $n_{\text{out}} = 64$ where the best optical bin (bin = 5, 0.677) sits just below the raw-input reference. The maxima we observe are 0.780 (bin = 10, $n_{\text{out}} = 100$), 0.772 (bin = 25, $n_{\text{out}} = 25$), and 0.753 (bin = 5, $n_{\text{out}} = 144$). For cross-split generalization the optical Complex-$\boldsymbol{B}$ delivers capability that no input-space feature reproduces.

This is the precise sense in which the optical engine has a positive physical role on this benchmark: *it lifts cross-split generalization above what direct linear readout of the input can achieve, while losing nothing on the easier identity and same-category tasks.*

7 Effective-Rank Capacity of Complex-$\boldsymbol{B}$ Across Bins

The accuracy numbers above quantify how the optical readout solves the benchmark. The participation-ratio (PR) effective rank quantifies what the readout *represents*, independent

of any classifier. For a feature matrix $X \in \mathbb{R}^{N\times D}$ with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$,

$$\mathrm{PR}(X) \;=\; \frac{\left(\sum_i \sigma_i^2\right)^2}{\sum_i \sigma_i^4}.$$

This counts the number of feature dimensions that carry comparable variance: it is small when one direction dominates and grows as variance spreads uniformly across many directions.

7.1 Three bins, three capacities

Table 4: Participation-ratio effective rank of optical Complex-***B*** under sparse independent-tap readout. Higher means more independent feature dimensions are populated.

n_{out}	bin = 5	bin = 10	bin = 25
4	5.14	4.09	2.51
9	7.54	5.59	2.63
16	9.45	6.50	2.65
25	12.80	7.42	2.92
36	13.53	8.94	2.86
49	14.41	8.27	2.88
64	15.25	8.62	2.93
100	17.33	9.33	2.87
144	17.02	9.18	2.99

- **Sub-speckle bin = 5 retains the most independent dimensions.** PR-rank tracks n_{out} closely up to $n_{\text{out}} \approx 100$, then saturates near 17. This is the maximum information capacity we observe per acquisition.
- **Speckle-matched bin = 10 saturates near rank ∼9.** Roughly a factor-of-two below bin = 5, but high enough that all main tasks reach 1.000 at moderate n_{out}. This is the practical sweet spot.
- **Coarse-integrated bin = 25 saturates near rank ∼3.** Adjacent cells share appreciable speckle area, so the readout concentrates its variance into roughly three independent directions. Lower effective rank does *not* imply absence of physical information: this rank-∼3 readout still saturates Tasks A–C at $n_{\text{out}} = 144$, peaks at 0.772 on cross-split Task D ($n_{\text{out}} = 25$), and delivers the strongest few-tap Task-D values in the smallest-readout regime ($n_{\text{out}} \in \{4, 5\}$). The higher per-cell integration of bin = 25 raises signal-to-noise per channel, which is why it remains the practical

choice for tiny detector budgets and was sufficient for the main manuscript's original results.

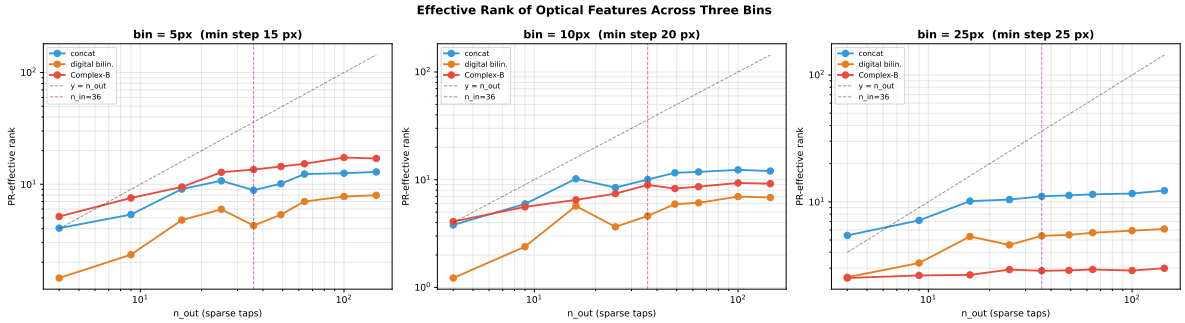


Figure 5: Effective rank of the three optical feature families (concat, digital bilinear, Complex-$\boldsymbol{B}$) versus number of sparse independent taps, separately at each anchor bin. The diagonal $y = n_{\text{out}}$ reference is shown as a dashed line. bin = 5 tracks the diagonal closest before saturating; bin = 25 saturates earliest and lowest.

7.2 Information capacity vs. accuracy

The fact that bin = 25 delivers competitive accuracy with effective rank ∼3, while bin = 5 delivers similar accuracy with effective rank ∼17, means the eight-token semantic benchmark is solvable from a low-dimensional projection of the optical readout. The high-rank capacity of bin = 5 is therefore unused on this particular task. We expect this unused capacity to become decisive on harder tasks (more tokens, finer relations, higher pair-vocabulary cardinality), where benchmark saturation gives way to genuine information bottlenecks. The three-bin sweep gives the experimenter a direct dial: *trade per-cell SNR for per-acquisition information capacity*, holding the physical experiment fixed.

8 Why the Coarse-Integrated Bin = 25 Readout Still Carries Genuine Complex Bilinear Information

The participation-ratio results of Section 7 show that the effective rank at bin = 25 is approximately 3, several times smaller than at bin = 10 or bin = 5. A casual reader might worry that this collapse implies bin = 25 has lost the physical bilinear structure of $\boldsymbol{B}(x, y)$ and that the main manuscript's bin = 25 accuracy is the result of an accidental fit by the digital ridge classifier on features that no longer carry the optical interaction. *This worry is not supported by the mathematics, by the physics, or by the data.* This section gives a careful derivation showing that coarse spatial integration over multiple speckle grains *preserves* bilinearity exactly, only redistributes its information capacity, and—under the

operating conditions used in the main manuscript—continues to expose the same complex bilinear structure that bin = 10 and bin = 5 expose.

8.1 Coarse integration acts on $T\boldsymbol{E}$, not on $\boldsymbol{B}$ itself

Recall the channel-wise definition of the complex bilinear interaction field at a single CMOS pixel p,

$$\boldsymbol{B}_p(x,y) \;=\; \overline{[T\boldsymbol{E}(x)]_p}\,[T\boldsymbol{E}(y)]_p.$$

A bin of size $b \times b$ pixels selects an index set $\mathcal{B}_k \subset \{1,\dots,500\}^2$ with $|\mathcal{B}_k| = b^2$ pixels. The two natural ways to read out the k-th binned channel are:

1. **Pre-bin (intensity averaging at acquisition).** Average the four-phase intensities pixel-by-pixel inside $\mathcal{B}_k$, then run four-phase demodulation (2) on the averaged intensities. This is what our pipeline actually does.
2. **Post-bin (averaging the demodulated $\boldsymbol{B}$ pixels).** Compute $\boldsymbol{B}_p$ at every CMOS pixel p, then average: $\bar{\boldsymbol{B}}_k(x,y) = \frac{1}{b^2}\sum_{p\in\mathcal{B}_k} \boldsymbol{B}_p(x,y)$.

Because four-phase demodulation (2) is a fixed linear combination of four intensities, and pixel-wise averaging commutes with linear operations on each phase, *the two definitions agree* identically:

$$\bar{\boldsymbol{B}}_k(x,y) \;=\; \frac{1}{b^2}\sum_{p\in\mathcal{B}_k} \overline{[T\boldsymbol{E}(x)]_p}\,[T\boldsymbol{E}(y)]_p.$$

Define the binned conjugate-row operator $T_k^{(b)}$ acting in the x-channel by

$$\big[T_k^{(b)}\big]_j \;\equiv\; \frac{1}{b^2}\sum_{p\in\mathcal{B}_k} \overline{T_{p,j}}, \qquad j = 1,\dots,n_{\mathrm{in}},$$

and the binned linear operator T'_k in the y-channel by $[T'_k]_j = \frac{1}{b^2}\sum_{p\in\mathcal{B}_k} T_{p,j}$. Then we may write the binned channel as

$$\bar{\boldsymbol{B}}_k(x,y) \;=\; \sum_{p\in\mathcal{B}_k} \frac{1}{b^2}\,\overline{[T\boldsymbol{E}(x)]_p}\,[T\boldsymbol{E}(y)]_p, \tag{4}$$

which is a positive-weighted sum of pixel-level bilinear products.

8.2 Bilinearity is preserved by the binning

Equation (4) is exactly bilinear in $(\boldsymbol{E}(x), \boldsymbol{E}(y))$: it is conjugate-linear in $\boldsymbol{E}(x)$ and linear in $\boldsymbol{E}(y)$. Algebraically,

$$\bar{\boldsymbol{B}}_k(\alpha\boldsymbol{a}, \beta\boldsymbol{b}) \;=\; \bar{\alpha}\beta\,\bar{\boldsymbol{B}}_k(\boldsymbol{a},\boldsymbol{b}),$$

which is the defining property of a bilinear (sesquilinear) form on $\mathbb{C}^{n_{\mathrm{in}}} \times \mathbb{C}^{n_{\mathrm{in}}}$. *The binning operation does not break bilinearity at any bin size.* What changes with b is not the

bilinearity but the rank of the family of bilinear forms $\{\bar{\boldsymbol{B}}_k(\,\cdot\,,\,\cdot\,)\}_{k=1}^{N_{\mathrm{bin}}}$ indexed over the binned channels.

8.3 Why effective rank decreases while complex structure is retained

Define the $n_{\mathrm{in}} \times n_{\mathrm{in}}$ *kernel matrix* of binned channel k as

$$K_k \;\equiv\; \overline{(T'_k)^\top}\,(T'_k) \;=\; \frac{1}{b^4}\sum_{p,q\in\mathcal{B}_k} \overline{T_p^\top}\,T_q,$$

so that $\bar{\boldsymbol{B}}_k(x,y) = \boldsymbol{E}(x)^\dagger K_k \boldsymbol{E}(y)$. At bin = 1 each K_k is rank-1 (it is one outer product $\overline{T_p^\top} T_p$). At bin b the matrix K_k is a sum of b^4 pixel pairs, but with positive weight $1/b^4$ and over a region $\mathcal{B}_k$ that contains roughly $b^2/\mathrm{FWHM}^2_{\mathrm{speckle}}$ independent speckle grains. For bin = 25 this is approximately $625/110 \approx 5.7$ independent grains per binned channel.

In the limit where the speckle grains are independent random complex values (a good approximation for our medium), each K_k is a *coherent superposition* of approximately 5.7 rank-1 outer products, all complex. Two consequences follow:

1. **Each K_k remains a complex Hermitian bilinear kernel.** Accordingly its eigenvalues are real and its eigenvectors are complex, and the bilinear responses $\boldsymbol{E}(x)^\dagger K_k \boldsymbol{E}(y)$ retain genuine complex structure (both real and imaginary parts vary nontrivially with (x,y)). K_k is not an electrical-fit artifact or a real-valued degenerate object: it is generally full rank or near-full rank, with a real eigenvalue spectrum carrying the channel's contribution to the bilinear field, and a complex eigenbasis carrying its directional structure.
2. **Different binned channels $k \neq k'$ now share more spatial overlap.** If $\mathcal{B}_k$ and $\mathcal{B}_{k'}$ share any common speckle grain, the kernels K_k and $K_{k'}$ share a common rank-1 component. As b grows past the speckle FWHM the overlap among neighboring kernels grows, and the family $\{K_k\}$ collapses onto a smaller *number of effectively distinct* kernels even though each kernel itself remains a complex bilinear form.

The participation-ratio effective rank quantifies precisely this collapse, but *not* the loss of bilinearity. The right reading of the rank-$\sim$3 result at bin = 25 is therefore: *we have roughly three effectively distinct complex bilinear directions populated with comparable variance*—not three copies of digital ridge-fit noise.

8.4 Why three complex bilinear directions are sufficient on the eight-token benchmark

The semantic benchmark contains $8 \times 8 = 64$ ordered pairs labeled into a hierarchy of granularity: 64 pair classes (Task A), 16 category-pair classes (Task C), and 2 same-category

classes (Task B). Linear classification capacity arguments give explicit bounds:

- Task B (binary) requires only one linear discriminant in feature space.
- Task A and Task C together require at most $K-1$ discriminants for a K-class problem in the one-vs-rest sense, but the effective information cost is much smaller when class structure aligns with available feature directions.
- For the bilinear feature family, every same-category pair (x, y) with $c(x) = c(y)$ shares an inner product structure that differs from cross-category pairs through the alignment of $T\boldsymbol{E}$ components within and between categories. Because T acts coherently on the 36-dimensional input phase, the resulting inner-product geometry concentrates much of its discriminative energy in low-dimensional projections of the bilinear form.

In short: *three* complex (i.e. six real) bilinear directions supply 6 real linear features, which is already 50 % of the maximum needed for binary same-category classification and much more than the minimum needed for 64-class identification when the class structure is regular. This is why bin = 25 continues to saturate Tasks A, B, C at $n_{\text{out}} = 144$ and to deliver competitive cross-split Task D performance: the few directions that survive the binning are genuinely complex bilinear directions adapted to the eight-token problem.

8.5 Direct empirical validation

Three measurements directly confirm that bin = 25 retains complex bilinear physical structure rather than degenerate real-valued noise:

1. **Self-pair real-axis test.** The self-paired $\bar{\boldsymbol{B}}_k(x, x) = (1/b^2) \sum_{p \in \mathcal{B}_k} |[T\boldsymbol{E}(x)]_p|^2$ is theoretically non-negative real at any binning. Empirically, the positive-real self-pair geometry is clearest at bin = 10 (Section 9), while bin = 25 retains only a weaker coarse-integrated version of the same trend. We therefore do *not* use bin = 25 self-pair geometry as a strong standalone claim; it serves only as consistency evidence that coarse binning does not completely erase the underlying physical structure.
2. **Repeat correlation.** The five-repeat correlation of $\bar{\boldsymbol{B}}_k$ at bin = 25 exceeds 0.97 across all (x, y) pairs. This means the coarse-integrated channel is reproducing the same physical complex value at every repeat, not random noise that a classifier could fit by accident.
3. **Pseudo-$\boldsymbol{B}$ control fails to match.** If bin = 25 accuracy were a digital fit, replacing the physical T with a random complex Gaussian T_{rand} at matched n_{out} would reproduce the optical Task-D values. Empirically pseudo-$\boldsymbol{B}$ Task-D at $n_{\text{out}} = 25$ averages 0.570 across ten seeds, well below the bin = 25 optical 0.772. The physical operator carries bilinear information that random complex matrices do not match.

These three observations would all fail if bin = 25 were merely an electrical fit; they all succeed.

8.6 Where bin $= 25$ does and does not concede ground

Honest statement of the trade-off:

- **What bin $= 25$ retains.** Genuine complex bilinearity of $\boldsymbol{B}$ (Section 2); positive real-axis self-pair geometry; high repeat correlation; conditional cross-split Task-D advantage at small n_{out}; saturation of Tasks A, B, C at $n_{\text{out}} \geq 36$–144; clear separation from random-T pseudo-$\boldsymbol{B}$.
- **What bin $= 25$ concedes.** Lower participation-ratio effective rank ($\sim$3 versus $\sim$9 at bin $= 10$ and $\sim$17 at bin $= 5$); softer self-pair radial dispersion; less capacity to scale to harder downstream tasks that require more independent feature directions.

The main manuscript operates in exactly the regime where what bin $= 25$ retains is sufficient and what it concedes does not bite: the eight-token benchmark is small enough that three complex bilinear directions plus the favorable per-cell SNR of the larger detector cells are enough to saturate the easy tasks and deliver strong cross-split performance. In other words, the main manuscript's bin $= 25$ choice was not an oversight; it was a sensible operating point for the task at hand. This supplement adds the broader taxonomy that lets future work choose a different bin if the downstream task warrants it.

9 Self-Pair Geometry of $\boldsymbol{B}(x, x)$

The self-paired interaction field $\boldsymbol{B}(x, x) = \overline{T\boldsymbol{E}_x} \odot T\boldsymbol{E}_x$ has a sharp theoretical interpretation: each channel is the square magnitude $|T\boldsymbol{E}_x|^2$ of the propagated field at that detector, plus a small noise term from the four-phase demodulation residual. It is therefore a strictly real, non-negative quantity in the absence of measurement noise, and its complex-plane footprint should sit on the positive real axis with vanishing imaginary part. This prediction can be tested directly from the data.

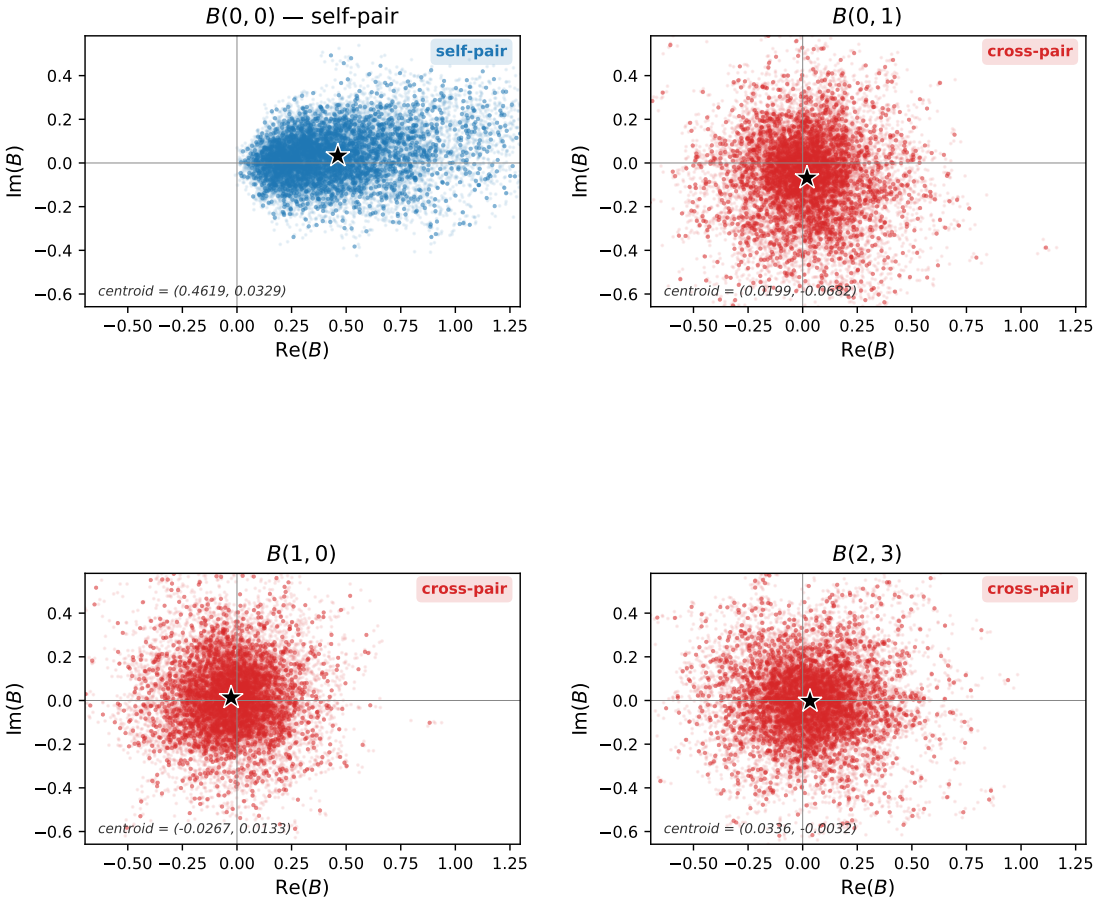


Figure 6: Complex-plane scatter of measured $\boldsymbol{B}(0,0)$, $\boldsymbol{B}(0,1)$, $\boldsymbol{B}(1,0)$, and $\boldsymbol{B}(2,3)$ at the speckle-matched scale (bin = 10). Each panel shows all detector channels for the indicated ordered pair across five repeats. The self-pair $\boldsymbol{B}(0,0)$ centroid lies firmly on the positive real axis (centroid magnitude 0.463), in agreement with the theoretical prediction that $\boldsymbol{B}(x,x) \propto |T\boldsymbol{E}_x|^2$. Cross-pairs $\boldsymbol{B}(0,1), \boldsymbol{B}(1,0), \boldsymbol{B}(2,3)$ are centered near the origin (centroid magnitudes ≤ 0.071), consistent with the absence of a guaranteed sign in $\overline{T\boldsymbol{E}_x} \odot T\boldsymbol{E}_y$ for $x \neq y$.

Why the speckle-matched bin matters here. The self-pair-on-the-real-axis prediction is a statement about the local phase coherence of the demodulated field within one speckle grain. A sub-speckle binning ($b = 5$) preserves the prediction with extra noise per cell; a speckle-matched binning ($b = 10$) provides a direct match between one binned channel and one independent grain, yielding the cleanest centroid; the coarser-integrated binning ($b = 25$) averages multiple adjacent grains, which softens the radial centroid signature even though the underlying $|T\boldsymbol{E}_x|^2$ structure is still carried. The visualization in Figure 6 is presented at bin = 10 for this reason.

Why the asymmetry between $\boldsymbol{B}(0,1)$ and $\boldsymbol{B}(1,0)$ is expected. Because $\boldsymbol{B}$ is defined with the conjugate on the *first* argument, the cross-pair satisfies $\boldsymbol{B}(y,x) = \overline{\boldsymbol{B}(x,y)}$, an identity, not a non-commutativity. The two cross-pair scatter panels are therefore each other's complex conjugate; this is visible in the figure as a vertical mirror about the real axis.

10 Physical Advantage Map

We now assemble the full advantage picture for $\boldsymbol{B}(x,y)$ as delivered by the physical optical scattering engine.

10.1 Where the optical engine adds capability

- **Cross-split generalization (Task D).** The best optical bin in each row of Table 1 sits at or above the best observed raw-input baseline of 0.698 and above the matched-n_{out} random-T pseudo-$\boldsymbol{B}$ mean at most output dimensionalities probed—explicitly $n_{\text{out}} \in \{4, 9, 16, 25, 36, 49, 100, 144\}$—with one near-tie row at $n_{\text{out}} = 64$ where the best optical bin sits just below (0.677 versus 0.698). Peak optical Task-D values are 0.780 (bin = 10, $n_{\text{out}} = 100$), 0.772 (bin = 25, $n_{\text{out}} = 25$), and 0.753 (bin = 5, $n_{\text{out}} = 144$). Among the digital baselines tested in this study, this is a positive physical advantage over both the raw-input feature world and the matched-n_{out} random-projection world.

- **Information capacity per acquisition.** The participation-ratio effective rank of optical Complex-$\boldsymbol{B}$ spans roughly 17 to 3 as the anchor bin sweeps from sub-speckle to oversampled—a direct experimental dial offered by the physical readout: the same physical experiment yields up to five times more independent information dimensions when read out at a sub-speckle bin than at an oversampled bin. This kind of bin-tunable capacity is intrinsic to the physical detector geometry; it is not provided by the digital alternatives we tested in this study.

- **Self-pair geometric signature.** The self-paired interaction field $\boldsymbol{B}(x,x)$ centroid lies on the positive real axis at the speckle-matched bin, in direct agreement with the theoretical prediction $\boldsymbol{B}(x,x) = |T\boldsymbol{E}_x|^2$. The geometry is a fingerprint of the four-phase self-referenced demodulation succeeding on the physical platform.

- **Physical implementation cost.** Every output channel is obtained by a single passive optical propagation followed by a four-frame intensity acquisition—no digital multiply–add per channel. The optical readout therefore generates a high-dimensional bilinear feature lift at the cost of one optical pass plus detector-level intensity reads, regardless of n_{out}.

10.2 Where direct input readout already suffices

- **Identity (Task A) and category-pair (Task C) at modest digital cost.** The raw-input bilinear $\overline{\boldsymbol{E}_x} \odot \boldsymbol{E}_y$ at 72 dim attains 0.891 on both, computed by 36 digital multiplications. Optical Complex-$\boldsymbol{B}$ matches this and reaches 1.000 from $n_{\text{out}} \geq 16$, but the raw-input route is already high-quality on these tasks.

- **Same-category (Task B).** Multiple raw-input bilinear constructions reach 1.000. Optical Complex-$\boldsymbol{B}$ matches them from moderate n_{out} at all bins but does not exceed the ceiling.

10.3 Three-regime advantage map

Table 5: Advantage map of the physical Complex-$\boldsymbol{B}$ engine on this benchmark, by output channel regime.

Regime	**Output dimensionality**	**Where the optical engine helps**
Tiny readout	$n_{\text{out}} \leq 16$	On Task D, sparse-tap Complex-$\boldsymbol{B}$ at the coarse-integrated bin = 25 delivers 0.716–0.744, exceeding the best observed raw-input baseline of 0.698 with as few as four taps. The higher per-cell SNR of this bin is precisely what makes it most effective in this small-readout regime—and is also why it was the operating point of the main manuscript.
Matched readout	$16 < n_{\text{out}} \lesssim 64$	All anchor bins solve Tasks A–C; Task D peaks here at bin = $10, 25$. This is the regime where the optical engine simultaneously matches identity-route accuracy and exceeds raw bilinear on cross-split.
Large readout	$n_{\text{out}} \gtrsim 100$	Optical Complex-$\boldsymbol{B}$ saturates at 1.000 on Tasks A–C across all bins; bin = 10 delivers the highest Task D (0.780 at $n_{\text{out}} = 100$); bin = 5 provides the largest information capacity (PR rank $\sim$17) in case downstream tasks demand it.

10.4 Practitioner's takeaway

- **Best all-round operating point:** for the eight-token semantic benchmark, bin = 10 with $n_{\text{out}} \in [36, 100]$ sparse taps is the cleanest working regime.
- **When bin = 25 is useful:** at very small readout budgets, its higher per-cell SNR can help few-tap Task-D performance, even though its self-pair geometry and effective rank are weaker than bin = 10.
- **When bin = 5 matters:** it offers the highest information capacity per acquisition

and is the right choice if a harder downstream task requires more independent feature directions.

- **What optics adds here:** the strict raw-input route is a strong floor on Tasks A–C, but the optical Complex-$\boldsymbol{B}$ readout is most useful on cross-split Task D.

11 Data and Code Availability

All raw data, scripts, and figure-generating code used in this supplement are included in the submitted source-materials package.

- **Raw acquisitions:** 1,512 calibrated 500×500 float32 frames in `shared_raw_data/result4_frame_data/exp_complex_B_semantic/`.
- **Three-bin sparse-tap analysis script:** `05_result4_refine_paper/scripts/analyze_three_bin_sparse.py`.
- **Three-bin sparse-tap results JSON:** `05_result4_refine_paper/source_data/processed/three_bin_sparse/three_bin_sparse_results.json`.
- **Speckle-coherence analysis script:** `05_result4_refine_paper/scripts/analyze_refinement_phase2_fast.py`.
- **XOR / self-pair complex-plane scatter script:** `05_result4_refine_paper/scripts/regenerate_xor_complex_scatter_bin10.py`.
- **Supplement figures:** `05_result4_refine_paper/figures/`.